Task 1: Data Cleaning and Preprocessing

Mall Customer Segmentation Dataset

Summary

The Mall Customer Segmentation dataset includes details like Customer ID, Age, Gender, Annual Income and Spending Score. By analysing this type of dataset helps in understanding customer behaviour, tastes and preferences which eventually support adopting effective marketing strategies.

Data Loading and Intial Overview

```
# Loading the Dataset
import numpy as np
import pandas as pd
df = pd.read_csv(r"C:\Users\DHANYA\Downloads\archive (5).zip")
print("Dataset is loaded successfully!")
```

```
Dataset is loaded successfully!
```

```
df
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

```
#  Display Column Information and Non-Null Counts
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

Data Pre-processing

```
# Checking missing values
df.isnull().sum()
```

```
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64
```

Note

- So it is confirmed that the given dataset is free from missing values.

```
# Droping Duplicate Values
df.drop_duplicates()
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

```
# Checking duplicated values
df.duplicated().sum()
```

```
np.int64(0)
```

Note

- This means that the dataset is free from duplicates.

```
# Checking Data Types
df.dtypes
```

```
CustomerID                 int64
Gender                    object
Age                        int64
Annual Income (k$)         int64
Spending Score (1-100)     int64
dtype: object
```

Note

- The dataset is clean with appropriate data types - numeric features for analysis and on categorical variable (Gender) for grouping.

Export as CSV

```
# Exporting Cleaned Dataset
df.to_csv("cleaned_mall_customer_segmentation_dataset.csv", index=False)
print("CSV file successfully exported!")
```

```
CSV file successfully exported!
```

Short Summary of Task:1

In Task 1, I performed essential data cleaning steps to prepare the dataset for analysis:

- Verified that the dataset has **no missing values**.
- Confirmed that there are **no duplicate records**.
- Checked and validated **data types**, ensuring numeric features and one categorical variable (Gender) are correctly structured.
- Finally, I exported the **cleaned dataset**.

```
Start coding or generate with AI.
```